MILITARY SPECIFICATION
CHAFF ROVING BUNDLE (TYPE RR-( )/AL), GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the (Type RR-( )/AL) Chaff Roving Bundle. The chaff (radar reflector) roving bundle contains a component capable of creating confusing returns to radar systems operating within specified frequency ranges when used with the dispenser/cutter prescribe

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC: N3996

FSC-5865

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall
PPP-B-1055	Barrier Material, Waterproof Flexible
PPP-F-320	Fiberboard, Corrugated and Solid
TT-I-1795	Ink, Marking, Stencil, Opaque, "Porous and Non-Porous
QQ-S-781	Strapping, Steel, and Seals
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

Military

MIL-P-116	Preservation-Packing, Method of
MIL-B-131	Barrier Materials, Water Vaporproof, Flexible, Heat-Sealable
DoD-D-1000	Drawing, Engineering and Associated Lists
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment
MIL-I-43553	Ink, Marking, Epoxy Base
MIL-C-85691/1	Chaff Roving Bundle (Type RR-179/AL) Specification Sheet

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized Unit Loads
DoD-STD-480	Configuration Control-Engineering Changes, Deviations and Waivers

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-2071	Testing of Chaff Radar Cross-Section

2.1.2 Other Government documents, drawings and publications. The following Government documents, drawings, and other publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

DL 1050AS160	Chaff Roving Bundle (Type RR-179/AL)
DL 1050AS161	Chaff Roving Bundle, RR-179/AL Marking Preservation and Packaging of
DL 1050AS163	Shipping Crate, Chaff Roving Bundle
DL 1050AS165	Fixture, Vibration/Shock Roving Bundle

MANUALS

Naval Air Systems Command

NAVAIR 16-30ALE-43-1-1 Operation and Maintenance Manual for Chaff Countermeasure Dispensing Sets:
AN/ALE-43(V)1 PN 39-2250-001,
AN/ALE-43(V)2 PN 39-2260-001,
AN/ALE-43(V)3 PN 39-2270-001

NAVAIR 16-30ALE-43-2-2 Illustrated Parts Breakdown for Chaff Countermeasure Dispensing Sets:
AN/ALE-43(V)1 PN 39-2250-001,
AN/ALE-43(V)2 PN 39-2260-001,
AN/ALE-43(V)3 PN 39-2270-001

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

STANDARDS

American Society of Mechanical Engineers

ANSI Y14.5 Dimensioning and Tolerancing

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirement shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2, 6.2.1, and 6.3).

3.3 Chaff material. The chaff material shall be in accordance with the applicable specification sheet.

3.3.1 Rovings. Rovings shall meet the dimension and composition requirements of the applicable specification sheets. Rovings composed of individually coated filaments an shall have a dc resistivity of less than 10 ohms per linear inch.

3.3.2 Binder. When required by the applicable specification sheet, a binder coating shall be applied to the chaff material to aid in maintaining bundle integrity.

3.4 Design and construction. Each chaff roving bundle (see 6.4.1) including filaments, packaging, and packing; shall conform to all the requirements of this specification and applicable specification sheet for design, construction, and workmanship, except as otherwise specified in the contract (see 6.2.1). Configuration control shall be maintained in accordance with DoD-STD-480.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.1 Nomenclature and identification marking. Nomenclature and identification marking shall be in accordance with MIL-STD-129, MIL-STD-130, and as specified herein.

3.4.2 Item definition. The chaff roving bundle covered by this specification shall consist of a continuous strand of metalized glass filaments as specified in the applicable specification sheet.

3.4.3 Physical characteristics. The chaff roving bundle shall be functionally compatible with the AN/ALE-43 (see 6.4.2) Dispensing Set. Neither the operational reliability of the dispenser nor the effectiveness of the chaff shall be degraded when the chaff roving bundle is used with this dispenser.

3.5 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions.

a.	Temperature	Room ambient (25 degree +/- 10 degree C)
b.	Altitude	Normal Ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity

3.6 Service conditions. The chaff roving bundle shall operate as required under any of the environmental service conditions or combinations of these conditions specified in MIL-STD-810 as modified herein. Tolerances shall be in accordance with the general requirements of MIL-STD-810.

3.7 Performance requirements. Unless otherwise specified herein, values set forth to establish the specified performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth will be specified herein.

3.7.1 Performance characteristics. Performance characteristics shall be in accordance with MIL-STD-2071 and the applicable specification sheet, when the chaff roving bundle is tested in accordance with the requirements specified herein.

3.8 Detail requirements. The chaff roving bundle contents shall meet the design, construction, and physical requirements specified herein. Drawings shall be interpreted in accordance with DoD-D-1000 and ANSI Y-14.5.

3.8.1 Chaff roving bundle.

3.8.1.1 Function. Each individual chaff roving bundle shall be of the type compatible with the AN/ALE-43(V) dispenser(s), (see 3.4.3). The roving bundle shall be formed by winding aluminum coated glass filaments into

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

right cylinders in a manner which allows the roving to be withdrawn from the core of the package.

3.8.1.2 Form factor. The chaff roving bundle payload (see 6.4.7) shall have diameter, height and rim thickness requirements as specified in the applicable specification sheet.

3.8.1.3 Outer wrap. The roving bundle shall be covered with an open-ended, protective, plastic film prior to packaging in accordance with the applicable DL.

3.8.1.4 Marking of outer wrap. The roving bundle outer wrap shall be marked in accordance with MIL-STD-130 indicating "chaff type, National Stock Number (NSN), including Navy Ammunition Code Number (NALC), part number, contract vendors number and lot number". The outer wrap itself may be stamped with waterproof, petroleum resistant ink or labels may be used.

3.8.1.5 Chaff roving bundle components. A chaff roving bundle shall consist of the components specified in the applicable DL and specification sheet.

3.8.1.6 Dipole configuration. The roving material as contained in the roving bundle shall be capable of being cut into specified dipole lengths with not greater than of 5 percent of entanglement or incomplete cuts. Final judgment of entanglement or incomplete cuts shall be made after the functional test. A visual judgement shall be used to determined if 5 percent of the dipoles are not being cut or breaking at a point not intended. If this is not discernible from measurements, the RCS data may assist in the final judgement.

3.8.1.7 Weight. The minimum gross weight of a chaff roving bundle shall be as specified in the applicable specification sheet.

3.9 Workmanship. The Chaff Roving Bundle, including all parts shall be manufactured in accordance with DL 1050AS160, the applicable specification sheet and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government will conduct all flight tests.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspections (see 4.2)

b. Initial Production Inspection (see 4.3)

c. Quality Conformance Inspections (see 4.4)

4.1.3 Lot information. A lot shall consist of a group of items of a specific configuration; produced within the same time frame at one location; by the same production line and production technique; and submitted for acceptance, inspection, or use at one time.

4.2 First article inspection. First article inspection shall be conducted by the contractor on 11 chaff roving bundles representative of the production units to be supplied under the contract. These chaff roving bundles shall be inspected according to the approved first article inspection procedure (see 4.5). No first article inspections shall be conducted prior to acceptance of the first article test procedures by the procuring activity. First article samples shall not be submitted for functional (Government) tests until in-house tests have been conducted in accordance with the approved test procedures, and the test results have been accepted by the contracting officer (see 6.2.2).

4.2.1 Scope of inspections. Three roving bundles shall be selected for first article inspections (see Table I) to verify that the chaff roving bundles meet all the requirements of this specification, the applicable specification sheet, and the contract. In addition, one dispenser load (eight chaff roving bundles) shall be selected for first article dispenser compatibility tests as specified herein and in the applicable specification sheets (see 3.4.3 and 3.8.1.1). The tests on each bundle shall be performed in the order specified in Table I. The bundle number shall be clearly indicated on each chaff roving bundle before first article tests are started.

TABLE I. First article inspections - schedule and sequence

Inspection	Requirement Paragraph	Inspection Paragraph	Chaff Roving Bundle			
			1	2	3	4-11
Chaff Materials	3.3	4.6	X			X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspections - schedule and sequence.- continued

Chaff Roving Bundle	3.8.1	4.7	X	X	X	X
High Temperature	3.6	4.8.1		X		
Low Temperature	3.6	4.8.2		X		
Temperature Shock	3.6	4.8.3		X		
Humidity	3.6	4.8.4		X		
Shock	3.6	4.8.5			X	
Vibration	3.6	4.8.6			X	
Functional	3.7	4.9	X	X	X	
Compatibility	3.8.1.1	4.10				X

4.2.2 First article approval. Approval of the first article sample will be by the procuring activity upon satisfactory completion of all inspection. No production chaff roving bundles shall be delivered prior to the approval of the first article sample.

4.2.3 Production chaff roving bundles. Chaff roving bundles supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each chaff roving bundle shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of noncompliance with the above shall constitute cause for rejection, and it shall be the obligation of the contractor to make necessary correction as approved by the procuring activity.

4.3 Initial production inspections. One of the first 20 production roving bundles shall be selected and sent at the contractor's expense to a designated Government laboratory for inspections. This sample shall be selected by the Government representative after it has successfully passed all individual inspections. No other inspections shall be conducted on the sample prior to starting the initial production inspections. The first article samples or units originally prepared for first article shall not be selected for this inspection.

4.3.1 Scope of inspections. The sample may be subjected to any and all inspections the procuring activity deems necessary to assure that the production equipment is equal to the previously approved sample in design, construction, workmanship, performance, and quality, and that it meets all requirements specified herein and in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.3.2 Initial production sample approval. Approval of the initial production sample will be by the procuring activity upon satisfactory completion of all inspections. Any design, material, or performance defect made evident during this inspection shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the inspections shall be cause for deliveries under the contract to cease until proper corrective action is approved and accomplished.

4.4 Quality conformance inspections. The contractor shall furnish all samples and, with the exception of Government conducted flight tests, shall be responsible for accomplishing the quality conformance inspections. All examinations and tests shall be under the supervision of the Government representative. When specified in the contract or purchase order, the contractor shall prepare inspection reports showing quantitative results for all applicable quality conformance inspections (See 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual Inspections (See 4.4.1)

b. Sampling Inspections (See 4.4.2)

c. Special Inspections (See 4.4.3)

4.4.1 Individual inspections. Individual inspections are those inspections conducted on each chaff roving bundle. Individual inspections for the chaff roving bundles shall be as specified in Table II. Failure to pass any of these inspections shall be cause for rejection of the chaff roving bundle.

4.4.2 Sampling inspections. Chaff roving bundles selected for sampling inspections shall first have passed the individual inspections. Chaff roving bundles shall be selected for sampling inspections by the Government representative in accordance with the schedule specified herein.

TABLE II. Individual inspections

Inspection	Requirement Paragraph	Inspection Paragraph
Form Factor	3.8.1.2	4.7.1
Retainer	3.8.1.3	4.7.2
Marking of Retainer	3.8.1.4	4.7.3
Chaff Roving Bundle	3.8.1.5	4.7.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

TABLE II. Individual inspections.- continued

Packaging and Packing	5.3, 5.4	4.7.6

4.4.2.1 Scope of inspections. Sampling inspections shall include examinations and tests for verification of manufacturing and performance requirements.

4.4.2.1.1 Sampling inspection for verification of manufacturing processes. Sampling inspections for verification of the manufacturing processes shall include, as a minimum, those in-process inspections specified in 4.6 and 4.7.

4.4.2.1.2 Sampling inspections for verification of performance requirements. Sampling inspections for verification of performance requirements (Government conducted flight tests) shall be conducted on chaff roving bundles to demonstrate conformance to the requirements of 4.9 and the applicable specification sheet when conducted in accordance with the procedures of MIL-STD-2071. Sampling shall be performed to provide two roving bundles from each production lot for inspection.

4.4.3 Special inspections. Special inspections shall be conducted on three chaff roving bundles for the purpose of checking the effect of any design or material change on performance, and to assure adequate quality control. The chaff roving bundles selected for special inspections may be selected from those previously subjected to sampling inspections.

4.4.3.1 Special inspection schedule. Selection of chaff roving bundles for special inspections shall be made as follows:

a. On early chaff roving bundles after an engineering or major change.

b. Whenever failure reports or other information indicate that additional inspections are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of inspections. Special inspections shall consist of such examinations and tests as specified in the contract (see 6.2.2). Inspection procedures previously approved for the first article inspection shall be used where applicable.

4.4.4 Chaff roving bundle. Should a failure occur during either the sampling or special inspections, the following action shall be taken (see 6.2.2):

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

c. Where practical, include an inspection in the individual inspections to check all units for this requirement until assurance is obtained that the defect has been corrected.

4.5 Inspection procedures. The right is reserved by the procuring activity or the Government inspector to modify the inspections or require any additional inspections deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of inspection procedures. When approved inspection procedures are available from previous contracts, such procedures may be used provided their use is acceptable to the procuring activity; however, the right is reserved by the procuring activity to require modification of such procedures, including additional inspections, when deemed necessary (see 6.2.1 and 6.2.2).

4.6 Inspection of materials.

4.6.1 Chaff material inspection. Examinations of the materials receiving inspection report for each receiving inspection lot (see 6.4.10), and on-line inspection of chaff roving lots (see 6.4.9) as specified herein shall be made to adequately demonstrate compliance with the requirements of this specification and the applicable specification sheet.

4.6.1.1 Chaff roving inspections.

4.6.1.1.1 Dipole characteristics inspection. Not less than, a ten foot length of roving from each chaff roving lot may be weighed, as a unit, to verify the in-process weight requirement.

4.7 Chaff roving bundle inspections.

4.7.1 Form factor. Chaff roving bundle weight and dimensional examination shall confirm compliance with 3.8.1.2.

4.7.1.1 Payload dimensions. Repeatable mechanical measurements shall be made to verify the payload rim thickness, height and diameter to verify the requirement of the applicable specification sheet.

4.7.1.2 Chaff roving bundle. The gross weight of the chaff roving bundle shall be measured to verify the requirement of the applicable specification sheet.

4.7.2 Outer wrap. Visual examination of the bundle shall be made to confirm compliance with 3.8.1.3.

4.7.3 Marking of outer wrap. Visual examination of the bundle shall be made to confirm compliance with 3.8.1.4.

4.7.4 Chaff roving bundle components. Verification of the presence of all components listed in applicable DL and the applicable specification sheet shall be made through visual inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.7.5 Dipole configurations. Visual examination of roving cuttings shall be made to confirm compliance with 3.8.1.6.

4.7.6 Packaging and packing. Visual examination shall be made of the packaging, packing and markings to assure compliance with the requirements of Section 5 of this specification. Any discrepancies from the specification shall be cause for rejection.

4.8 Environmental tests. The simulated environmental tests specified below shall be performed in the order presented herein. Tolerances for these tests shall be in accordance with MIL-STD-810 unless otherwise specified herein. The Chaff Roving Bundles shall not show signs of appreciable damage or degradation, and shall meet all dimensional and performance requirements after exposure.

4.8.1 High temperature. The chaff roving bundle shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501.2, Procedure I, Category: Hot induced conditions including humidity.

4.8.2 Low temperature. The chaff roving bundle shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502.2, Procedure I. Temperature shall be maintained at not less than -54 degree C for a period of 24 hours after stabilization.

4.8.3 Temperature shock. The chaff roving bundle shall be subjected to a temperature shock test in accordance with MIL-STD-810, Method 503.2, except the diurnal cycle shall be deleted. Temperature is to cycle between -54 degree C and +71 degree C.

4.8.4 Humidity. The chaff roving bundle shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.2, Procedure I, Cycle 1.

4.8.5 Shock. The chaff roving bundle shall be subjected to a shock test in accordance with MIL-STD-810, Method 516,3, Procedure I.

4.8.5.1 Shock pulse. The pulse shape shall be as shown on Figure 516.3-1 of MIL-STD-810 except that an Electrodynamic Vibration machine maybe used, and if so, the tolerances of the specified pulse maybe modified to allow amplitude variation up to +/-20 percent of maximum amplitude prior to the initiation of the sawtooth ramp. All other tolerances apply. (The change in tolerance allows for the normal necessary pre- and post-pulse for Electrodynamic Machine Shock Excitation.)

4.8.5.2 Shock fixture. The shock fixture shall conform to DL 1050AS165. There shall be a control accelerometer for each axis as specified in DL 1050AS165.

4.8.6 Vibration. The chaff roving bundle shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.3, Procedure 7B; except as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

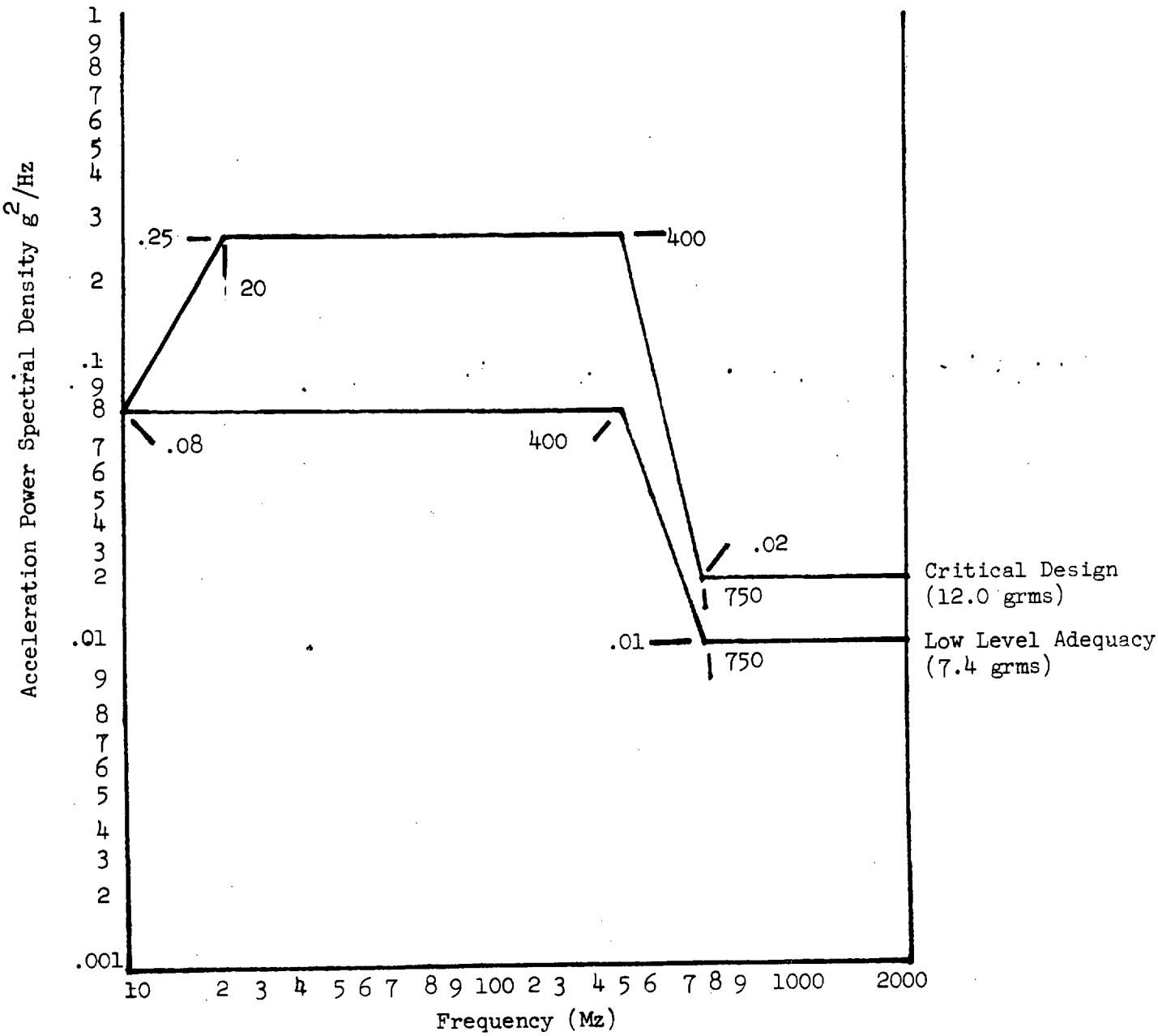


Figure 1. Low Level Adequacy and Critical Design Vibration Test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

a. The chaff roving bundle must withstand both a low level adequacy test and a critical design test.

b. Testing shall be performed in all three axis.

c. The same fixture used for the shock test shall be used for the vibration test.

d. The fixture shall be oriented in the flight-up position.

e. The control accelerometer location for each axis shall be as indicated on the vibration fixture drawing.

4.8.6.1 Vibration test sequence.

a. Low level adequacy vibration as shown on Figure 1 for 30 minutes in each axis.

b. Critical design vibration as shown on Figure 1 for one hour in each axis.

c. Repeat step a.

4.9 Functional test. All functional flight tests will be conducted by the Government in accordance with MIL-STD-2071, the applicable specification sheet, and as specified in 4.9.1 through 4.10.3.2.

4.9.1 Flight tests. The Government performed flight tests will be conducted on first article, initial production, quality conformance, and special test units to verify performance capability. Tests will include Radar Cross-Section (RCS) test, in accordance with MIL-STD-2071 and as modified herein. Chaff/dispenser compatibility tests will be conducted in addition to RCS tests for first article chaff roving bundles.

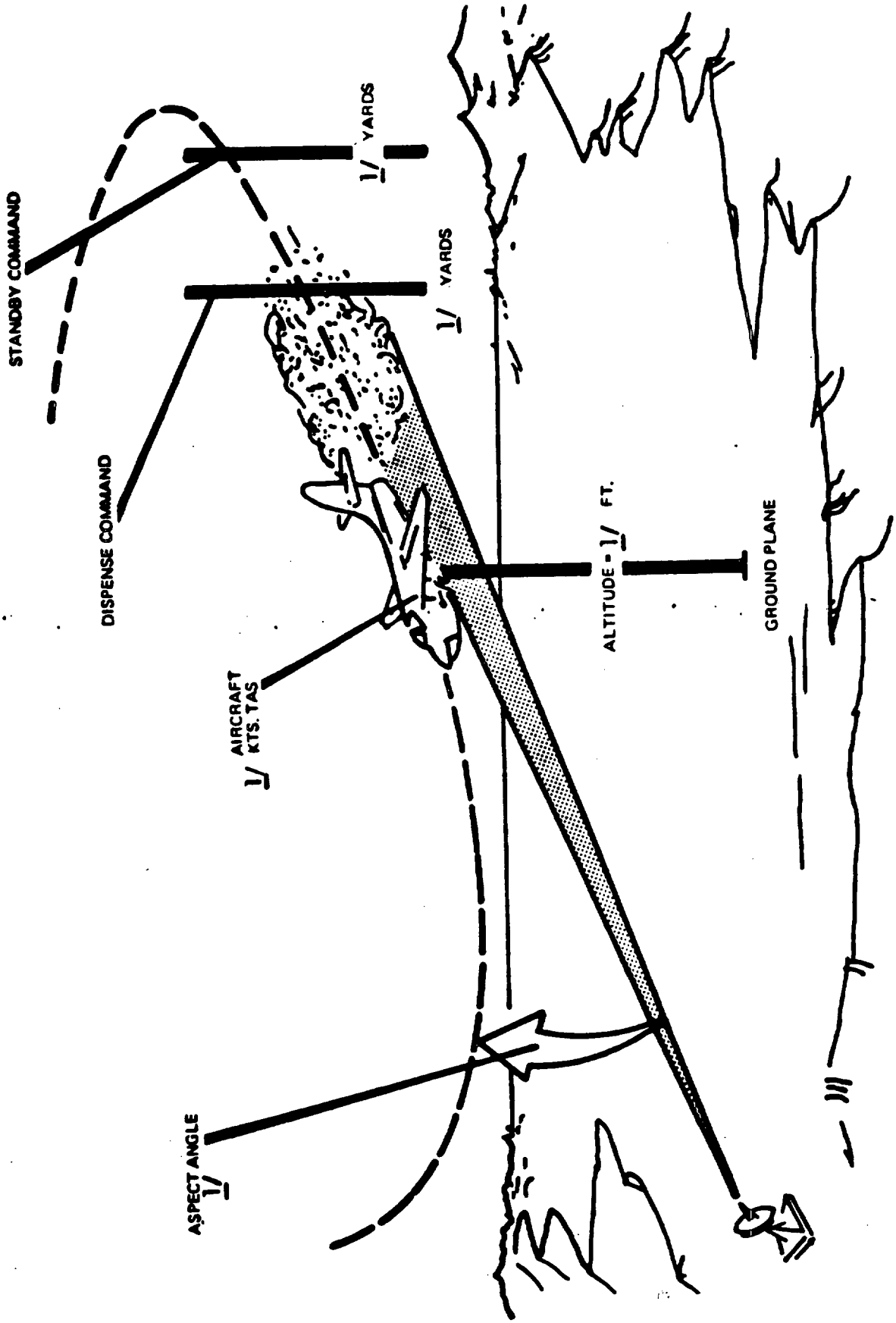
4.9.1.1 Aircraft. Flight tests will be conducted by the Government using a Navy aircraft. For test purposes, the aircraft will maintain an altitude of the number of feet and a True Air Speed (TAS) of the number of knots specified in the applicable specification sheet (see 3.1).

4.9.1.2 Dispenser. The AN/ALE-43(V)1 dispenser shall be used for all RR-( )/AL flight tests. All tests will be conducted at continuous dispense rate of 300 inches per second (fixed) utilizing the cutter head as specified in the applicable specification sheet.

4.9.1.3 Loading. A single RR-( )/AL Chaff Roving Bundle will be loaded into the AN/ALE-43(V)1 for dispensing in accordance with the procedures of NAVAIR 16-30ALE-43-1-1 and the applicable specification sheet.

4.9.1.4 Radar. The RR-( )/AL chaff will be evaluated using a M-33 radar or approved equivalent radar equipped in accordance with the applicable specification sheet (see 3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.



1/ Refer to applicable specification sheet for flight profile data.

FIGURE 2. Flight profile for testing of the RR-( )/AL Chaff Roving

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.9.1.5 Sampling. The test objective will be to conduct 10 runs of 40 seconds duration each; however, minimum test requirements can be met with 10 seconds of dispensing per run, or combined total of 100 seconds of dispensing.

4.9.1.6 Special requirements, first article samples. First article samples at the completion of the test specified in 4.9.1.5 will be further tested for dispenser compatibility. Eight roving bundles will be loaded in a single AN/ALE-43(V) dispenser for dispensing. The samples will then be tested at altitudes, air speeds and dispense rates representative of operational usage to insure required performance of the complete chaff system.

4.9.2 Radar cross-section (RCS) procedures.

4.9.2.1 Aircraft operation. The aircraft will fly a "racetrack" course as shown on Figure 2. Chaff will be dispensed on the inbound radial. Upon initial radar acquisition, the aircraft will be directed to maximum range, then brought in on a preliminary run.

4.9.2.2 Data gathering. The test radar will track the aircraft during the entire radial inbound run. Provisions will be made to prevent breaklock by installing a radar beacon or Luneberg lens on the aircraft. Areas of old chaff inadvertently crossed, or segments of runs in which the radar broke lock on the aircraft will be identified using the shadow graph, and rejected as invalid data. As depicted on Figure 2, a standby and dispense command will be given at a range specified in the applicable specification sheet (see 3.1). Data recording will commence at the standby command and will continue for 10 seconds after termination of chaff dispensing. If old chaff creates an interference problem, a delayed start will be accepted. The delayed start, however, may result in a short run.

4.9.2.3 Data evaluation. Data will be evaluated to determine the presence of at least 100 seconds of pulse-to-pulse data meeting the requirements of the procedures for collecting data on chaff immediately behind the aircraft, and the requirements for aspect angle as specified in MIL-STD-2071.

4.9.2.4 Data processing. Data will be processed to obtain RCS values of dispensed chaff in the radar resolution cells, the number of feet behind the aircraft as specified in the applicable specification sheet. Processing will be accomplished in accordance with MIL-STD-2071. The data will be normalized to 10 degree aspect angle and 0 degree antenna boresight gain.

4.9.2.4.1 Data selection. Shadow graphs will be used to select the data to be processed. For each run, the first record of data containing good RCS values in Gate 10 will be used as the start of the RCS data for that run. Normally, the next 20 test samples will be processed. Test samples of 1024 pulses will be used. The test samples to be processed for each data run will be contiguous unless the radar errors necessitate breaking up the data samples in a run into two or three groups. In such cases, a total of 30 samples will be processed on each run if sufficient data are available. In all cases, data

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

will be processed only when the radar is tracking properly as shown by the shadow graph results, and in no case will a hole or weak spot in the chaff cloud (caused by poor chaff performance) be eliminated from the data to be processed.

4.9.2.4.2 Calibration check. RCS measurement of a standard calibration sphere will be made prior to any chaff measurements. The sphere RCS data will be compared to the theoretical RCS of the sphere, and the calibration constant (K) for the processor will be set to provide the measured sphere data as near as possible to the theoretical value of the sphere RCS. The corrected value of K will be used in the first and second level data processing.

4.9.2.4.3 Correction factors. Correction factors will be in accordance with the applicable specification sheet.

4.9.2.4.4 Statistical results. Results in the form of first level statistics will be accomplished by processing the selected data from shadow graph using the same 1024 pulses to obtain the RCS values for all the gates in each test sample. Shadow graphs will be used to define potential holes or weak spots in the chaff cloud, and when it is indicated that holes greater than those specified in 4.9.3.2 herein exist, the first level statistics of that portion of the data will be reprocessed in the following manner:

a. Test samples of 256 radar pulses will be used.

b. Reprocessing will start four test samples before the indicated hole in the first gate, and will continue two test samples beyond the hole as observed in the 10th Gate.

c. Various sections along the test sample may be processed to specify the exact severity of the holes or weak spots that are present.

4.9.3 Pass/fail criteria. The chaff roving bundles shall meet the requirements of 4.9.3.1 and 4.9.3.2 herein.

4.9.3.1 Radar cross-section (RCS). The average of the pulse-to-pulse RCS values as measured the number of feet specified in the applicable specification sheet behind the aircraft shall meet or exceed the number of square meters specified in the applicable specification sheet per gate for any unit to be acceptable. This average shall meet or exceed the number of square meters specified in the application sheet for at least 85 percent of the individual test runs.

4.9.3.2 Chaff cloud homogeneity. Discontinuities in the homogeneity of the chaff cloud will normally result in second level averages below those specified in 4.9.3.1. However, for acceptance purposes, discontinuities (holes) in the chaff cloud which are greater than 0.5 second as displayed on the shadow graph, and which average less than the 50 percent value of the first level processing of the run, will be cause for rejection of the chaff

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

roving bundle. The method for specifying the width and depth of the hole will be as specified in 4.9.2.4.4.

4.10 Chaff dispenser compatibility test. The procedures specified in 4.10.1 through 4.10.3.2 will be followed unless otherwise specified in the detailed test plan.

4.10.1 Flight profile. The flight profile will be specified in the detailed test plan. The altitude, air speed, and dispense rate combinations used will be representative of operational usage.

4.10.2 Data gathering for chaff/dispenser compatibility test. The test radars will track the aircraft during the entire inbound run. Areas of chaff dispensed on previous runs that are inadvertently crossed, or segments of runs in which the test radar breaks-lock on the aircraft, will be identified and assessed as to impact on the test results. If break-lock occurs, the radars shall be oriented on the centroid of the chaff cloud, and the test procedures will continue from that point in time. When the aircraft is successfully tracked to the cessation of dispensing, the radars will remain on the terminal portion of the chaff cloud for 10 seconds, after which the radars will orient to the centroid of the chaff cloud for one additional minute. RCS data recording will commence 10 seconds before chaff dispensing begins, and continue until 60 seconds after dispensing has ceased. Immediately following the flight test, the chaff dispenser will be removed from the aircraft and inspected in the laboratory. Inspection results will be entered on the log sheet as specified in MIL-STD-2071.

4.10.3 Pass/fail criteria for chaff dispenser compatibility test. The chaff roving bundle shall meet the requirements of 4.10.3.1 and 4.10.3.2 herein. A numerical value for chaff cloud RCS is not a part of the pass/fail criteria for this test, although this data should be maintained for data accumulation purposes.

4.10.3.1 Roving jams. Jams, not attributed to loading errors or dispenser failures, in excess for one roving bundle-jam per full pod (8 bundles) will be cause for rejection of the first article sample submission. A jam is considered as premature termination of dispensing chaff due to failure of the roving to feed properly.

4.10.3.2 Chaff cloud discontinuities (chaff dispenser compatibility test). Discontinuities in the homogeneity of the chaff cloud which are greater than those described in 4.9.3.2 herein will be cause for rejection of the first article sample.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order (see 6.2.1) in accordance with MIL-STD-794 and as specified herein. Unless otherwise specified in the contract or purchase order, the method of preservation-packaging shall be Method I,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

Submethod IA-15 in accordance with MIL-P-116 and this specification (see 6.2.1).

5.2 Packaging of chaff roving bundle in inner carton. One chaff roving bundle shall be packaged per inner carton.

5.2.1 Carton details. The inner carton shall be a Type CF, Class "Weather-Resistant", Variety SW, Grade V3C container (RSC) conforming to PPP-B-636 and DL 1050AS161. The bottom longitudinal closure of the carton shall be completed by using a suitable length of filament reinforced tape, Type II conforming to PPP-T-45, with a minimum width of three inches. The four end joints formed by the flaps shall be sealed with 60 pound Kraft paper tape with a minimum width of three inches. The top closure shall be completed after the carton is filled by using a suitable length of filament reinforced tape, Type II conforming to PPP-T-45, with a minimum width of three inches. A six inch length of this tape shall be left free at one end of the tape and folded back and sealed to itself for a length of three inches, to provide a handle for grasping to use the tape as a ripcord in unsealing the top longitudinal flaps. The four end joints formed by the flaps shall be sealed with 60 pound Kraft paper tape with a width of not less than three inches. Cartons shall conform to the test requirement specified herein.

5.2.2 Barrier material. The carton containing the chaff roving bundle shall be enclosed in a one-piece wrapping of water vapor proof, oil resistant barrier material, Class 1, conforming to MIL-B-131. The barrier material shall be vacate to form, and sealed by the application of heat as specified in MIL-B-131.

5.3 Packaging details.

5.3.1 Interior packaging. All inner cartons shall be packaged in accordance with MIL-P-116 and applicable DL.

5.3.2 Exterior packaging. Consignments shall be prepared for either domestic or overseas shipments as specified in the contract (see 6.2.1).

5.3.3 Marking. The inner carton and barrier material shall be permanently stamped in waterproof, petroleum resistant ink in accordance with MIL-STD-129 and applicable DL.

5.4 Packing (domestic and overseas packing of material in carton/crates). Twelve inner cartons per layer (two layers per crate) shall be placed in a wooden crate, Style A, B or I, constructed in accordance with PPP-B-601 and utilizing a crateliner composed of a Barrier Material, Waterproof, Flexible in accordance with PPP-B-1055.

5.4.1 Marking for shipments. Shipping containers shall be marked in accordance with 5.3.3 and as specified in the contract (see 6.2.1).

5.4.2 Palletizing. Palletizing shall be accomplished in accordance with applicable DL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Chaff roving bundles covered by this specification are intended for use in the AN/ALE-43(V) Dispensing Sets.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Chaff roving bundle type (see item description of applicable specification sheet).

c. Levels of preservation-packaging, packing and marking required (see Section 5).

d. If design, construction or workmanship are to be other than specified in the specification (see 3.4).

e. If first article is required (see 3.2, 6.3).

f. If test procedures are to be expanded or modified (see 4.5).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.5	Procedure, First Article Inspection	DI-T-4901	
4.2	Report, First Article Inspection	DI-T-4902	
4.4	Report, Production Inspection	DI-T-4904	
4.4.3.2, 4.5	Procedure, Production/Acceptance Inspection	DI-T-4903	
4.4.4	Failure Data Collection, Analysis and Corrective Action Plan	UDI-T-23719	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

(Data item descriptions related to this specification and identified in Section 6 will be approved and listed as such in DOD 5010.12L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample selected from the first 20 production item(s). The first article sample should consist of three chaff roving bundles. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waiver the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. The following definitions apply to this specification:

6.4.1 Chaff roving bundle. The chaff package configuration containing the chaff payload for use with the AN/ALE-43(V) dispensers.

6.4.2 AN/ALE-43(V). This nomenclature refers to an internal, high capacity, bulk chaff/dispenser system capable of fixed internal installation or pod mounting.

6.4.3 Chaff container. Houses the chaff supply in the form of roving bundles during dispensing operations applicable to AN/ALE-43(V)1 and (V)2 or 3.

6.4.4 Systems. As used herein refers to one AN/ALE-43 described above.

6.4.5 Pod. Pod is used to specify the structural case of the ALE-43 (V)1 and shall include the nose and tail cones.

6.4.6 Payload. The payload is the chaff carried in the chaff container.

6.4.7 Chaff cut. A quantity of dipoles cut from a roving to provide dipole elements capable of creating confusing radar signal returns.

6.4.8 Chaff filament. An assemblage of drawn metalized glass fibers produced by one machine prior to the roving assembly operation.

6.4.9 Lot, chaff roving. The chaff roving lot will contain the quantity of roving bundles specified in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

6.4.10 Lot, receiving inspection. A receiving inspection lot will consist of all material of one type received from one supplier at one time.

6.5 Preparing the chaff roving bundle for vibration and shock. The installation of the chaff roving bundle into the fixture is designed to simulate the actual mounting in the aircraft and/or pod. This mounting is fairly simple, but it does involve the use of a number of component parts. The following is included herein to facilitate and standardize the mounting method.

a. Equipment

(1) Chaff Roving Bundle

(2) Chaff Roving Bundle Vibration Fixture

b. Procedure

(1) Remove bundle from cardboard box

(2) Place bundle into the vibration fixture

(3) Remove cardboard insert from bundle core

(4) Unit ready for test

6.6 Subject term (keyword) listing.

1. AN/ALE-43(V)
2. Container, chaff
3. Cut, chaff
4. Filament, chaff
5. Lot, roving, chaff
6. Payload
7. Pod
8. Roving bundle, chaff
9. Systems

Preparing Activity:
Navy - AS
(Project 5865-N060)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.